



通信电子线路

Fundamentals and Design of Communication Circuits

主讲：李国兵



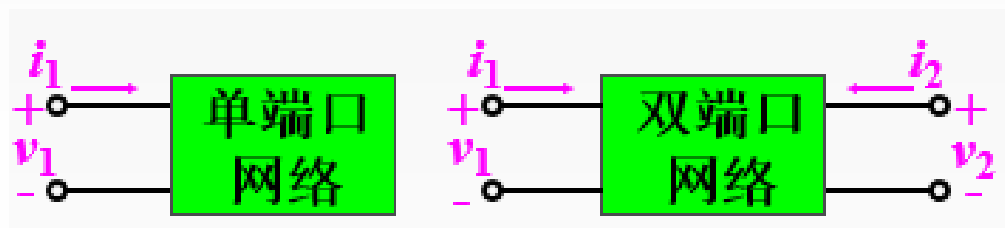
单端口和多端口网络

- 基本网络参量的含义
- 互联网络
- 网络特性及其应用
- 散射参量 (S参数)

4. 单端口和多端口网络

网络模型的作用

- ✓ 网络模型可以大量减少无源和有源器件数目；
- ✓ 避开电路的复杂性和非线性效应；
- ✓ 简化网络输入和输出特性的关系；
- ✓ 可以通过实验确定网络输入和输出参数，而不必了解系统内部的结构。



4.1 基本网络参量的含义

$$v_1 = Z_{11}i_1 + Z_{12}i_2 + \cdots + Z_{1N}i_N$$

$$v_2 = Z_{21}i_1 + Z_{22}i_2 + \cdots + Z_{2N}i_N$$

$\vdots$

$$v_N = Z_{N1}i_1 + Z_{N2}i_2 + \cdots + Z_{NN}i_N$$



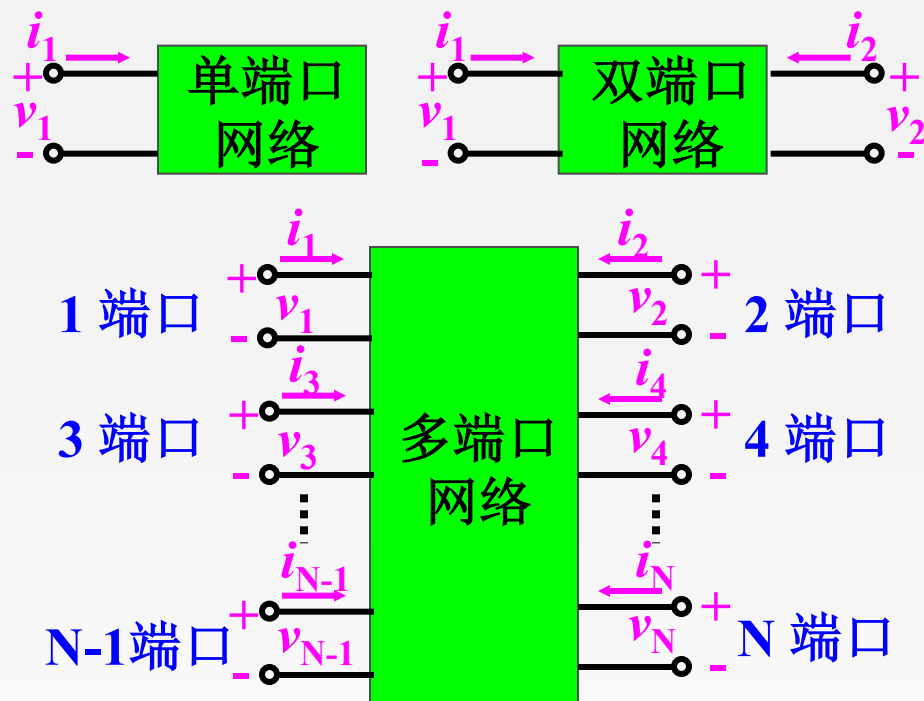
$$\begin{Bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{Bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1N} \\ Z_{21} & Z_{22} & \cdots & Z_{2N} \\ \vdots & \vdots & & \vdots \\ Z_{N1} & Z_{N2} & \cdots & Z_{NN} \end{bmatrix} \begin{Bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{Bmatrix}$$



$$\begin{Bmatrix} i_1 \\ i_2 \\ \vdots \\ i_N \end{Bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} & \cdots & Y_{1N} \\ Y_{21} & Y_{22} & \cdots & Y_{2N} \\ \vdots & \vdots & & \vdots \\ Y_{N1} & Y_{N2} & \cdots & Y_{NN} \end{bmatrix} \begin{Bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{Bmatrix}$$

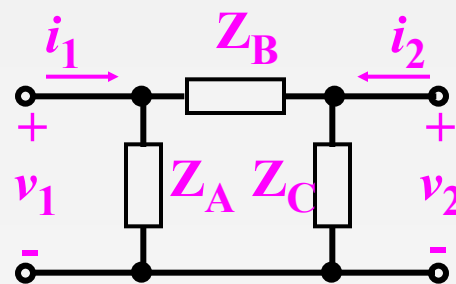
$$Z_{nm} = \left. \frac{v_n}{i_m} \right|_{i_k=0 \text{ (for } k \neq m)}$$

$$Y_{nm} = \left. \frac{i_n}{v_m} \right|_{v_k=0 \text{ (for } k \neq m)}$$



4.1 基本网络参量的含义

例4.1 求 π 形网络的阻抗矩阵和导纳矩阵。



解:
$$Z_{11} = \left. \frac{v_1}{i_1} \right|_{i_2=0} = \frac{Z_A(Z_B + Z_C)}{Z_A + Z_B + Z_C}$$

$$Z_{21} = \left. \frac{v_2}{i_1} \right|_{i_2=0} = \frac{Z_A Z_C}{Z_A + Z_B + Z_C}$$

$$Y_{11} = \left. \frac{i_1}{v_1} \right|_{v_2=0} = \frac{1}{Z_A} + \frac{1}{Z_B}$$

$$Y_{21} = \left. \frac{i_2}{v_1} \right|_{v_2=0} = -\frac{1}{Z_B}$$

$$Z_{12} = \left. \frac{v_1}{i_2} \right|_{i_1=0} = \frac{Z_A Z_C}{Z_A + Z_B + Z_C}$$

$$Z_{22} = \left. \frac{v_2}{i_2} \right|_{i_1=0} = \frac{Z_C(Z_A + Z_B)}{Z_A + Z_B + Z_C}$$

$$Y_{12} = \left. \frac{i_1}{v_2} \right|_{v_1=0} = -\frac{1}{Z_B}$$

$$Y_{22} = \left. \frac{i_2}{v_2} \right|_{v_1=0} = \frac{1}{Z_B} + \frac{1}{Z_C}$$

结论：通过假设网络端口为开路或短路状态，容易测得全部参数，且互易。

4.1 基本网络参量的含义

A矩阵(级联矩阵):

$$\begin{Bmatrix} v_1 \\ i_1 \end{Bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{Bmatrix} v_2 \\ -i_2 \end{Bmatrix}$$



h 参量矩阵(混合矩阵): $\begin{Bmatrix} v_1 \\ i_2 \end{Bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{Bmatrix} i_1 \\ v_2 \end{Bmatrix}$ 元素计算方法同前。



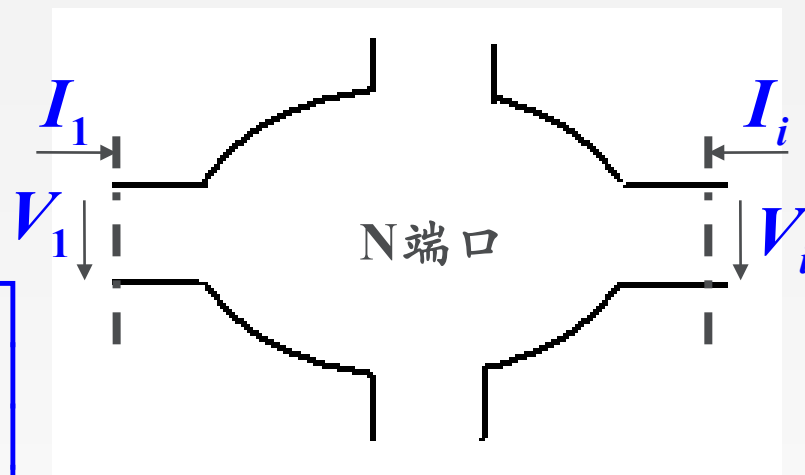
4.1 基本网络参量的含义



一、阻抗、导纳参量

端口参量: $[V]$ 、 $[I]$

$$\begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_N \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1N} \\ Z_{21} & Z_{22} & \cdots & Z_{2N} \\ \vdots & \vdots & \cdots & \vdots \\ Z_{N1} & Z_{N2} & \cdots & Z_{NN} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix}$$
$$\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} & \cdots & Y_{1N} \\ Y_{21} & Y_{22} & \cdots & Y_{2N} \\ \vdots & \vdots & \cdots & \vdots \\ Y_{N1} & Y_{N2} & \cdots & Y_{NN} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_N \end{bmatrix}$$



简写

$$[V] = [Z][I]$$

$$[I] = [Y][V]$$

关系

$$[Z] = [Y]^{-1}$$

网络参量: $[Z]$ 、 $[Y]$



4.1 基本网络参量的含义



$$Z_{ij} = \left. \frac{V_i}{I_j} \right|_{I_k=0, k \neq j}$$

其它端口开路

j 端口到 i 端口的转移阻抗(互阻抗)

$$Z_{ii} = \left. \frac{V_i}{I_i} \right|_{I_k=0, k \neq i}$$

其它端口开路,

i 端口的输入阻抗(自阻抗)

$$Y_{ij} = \left. \frac{I_i}{V_j} \right|_{V_k=0, k \neq j}$$

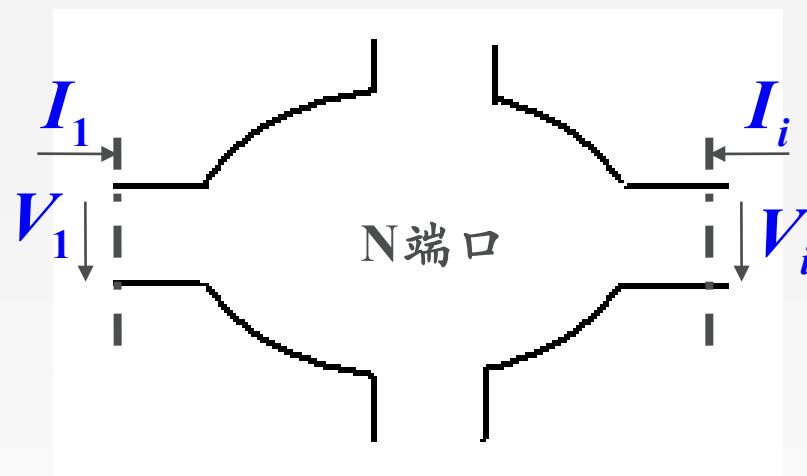
其它端口短路,

j 端口到 i 端口的转移导纳(互导纳)

$$Y_{ii} = \left. \frac{I_i}{V_i} \right|_{V_k=0, k \neq i}$$

其它端口短路,

i 端口的输入导纳(自导纳)



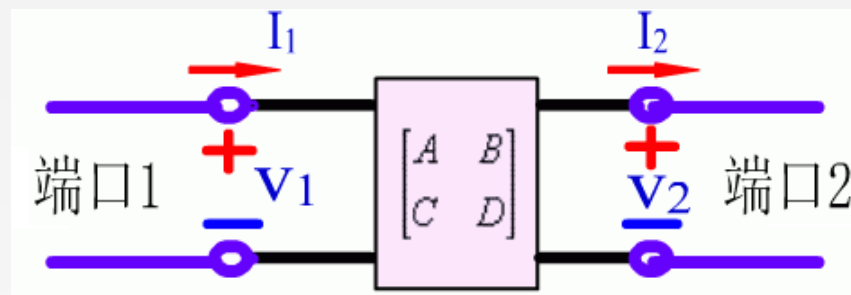


4.1 基本网络参量的含义



二、转移参量【A】

$$\begin{cases} v_1 = av_2 + bi_2 \\ i_1 = cv_2 + di_2 \end{cases} \quad \begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} v_2 \\ i_2 \end{bmatrix}$$



电流 i_2 的方向?

$$a = \left. \frac{v_1}{v_2} \right|_{i_2=0}$$

端口2开路, 2端口到1端口的电压转移系数

$$d = \left. \frac{i_1}{i_2} \right|_{v_2=0}$$

端口2短路, 2端口到1端口的电流转移系数

$$b = \left. \frac{v_1}{i_2} \right|_{v_2=0}$$

端口2短路, 2端口到1端口的转移阻抗

$$c = \left. \frac{i_1}{v_2} \right|_{i_2=0}$$

端口2开路, 2端口到1端口的转移导纳



4.1 基本网络参量的含义



三、混合参量【H】

$$\begin{bmatrix} v_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ i_2 \end{bmatrix}$$



$$h_{12} = \left. \frac{v_1}{v_2} \right|_{i_1=0}$$

端口1开路, 2端口到1端口的电压传输系数(反馈)

$$h_{21} = \left. \frac{i_2}{i_1} \right|_{v_2=0}$$

端口2短路, 1端口到2端口的电流传输系数(放大)

$$h_{11} = \left. \frac{v_1}{i_1} \right|_{v_2=0}$$

端口2短路, 端口1的输入阻抗

$$h_{22} = \left. \frac{i_2}{v_2} \right|_{i_1=0}$$

端口1开路, 端口2的输入导纳

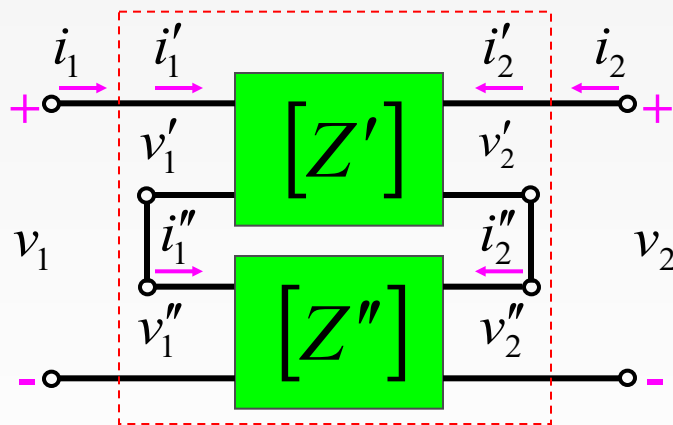
4.2 互连网络

4.2.1 网络的串联

每个电压相互叠加而电流不变则用Z参数：

$$\begin{Bmatrix} v_1 \\ v_2 \end{Bmatrix} = \begin{Bmatrix} v'_1 + v''_1 \\ v'_2 + v''_2 \end{Bmatrix} = [Z] \begin{Bmatrix} i_1 \\ i_2 \end{Bmatrix}$$

$$[Z] = [Z'] + [Z''] = \begin{bmatrix} Z'_{11} + Z''_{11} & Z'_{12} + Z''_{12} \\ Z'_{21} + Z''_{21} & Z'_{22} + Z''_{22} \end{bmatrix}$$

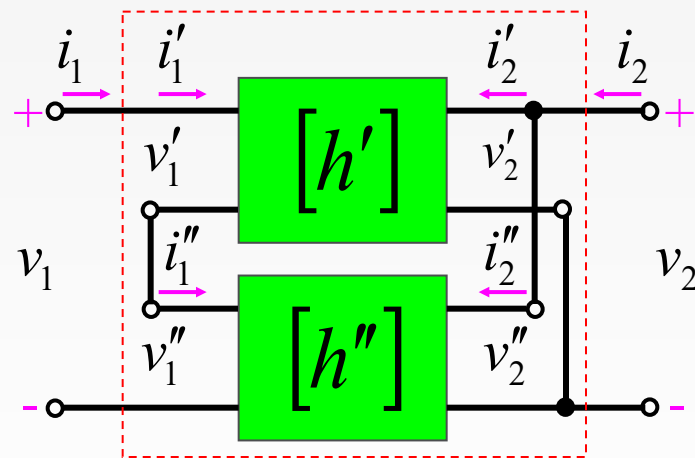


4.2 互连网络

4.2.1 网络的串联

若输入电压及输出电流叠加，而输入电流及输出电压不变则用 **h** 参数：

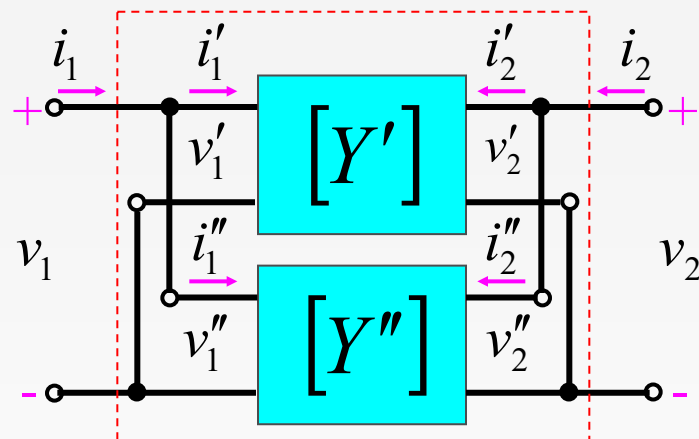
$$\begin{Bmatrix} v_1 \\ i_2 \end{Bmatrix} = \begin{Bmatrix} v'_1 + v''_1 \\ i'_2 + i''_2 \end{Bmatrix} = \begin{bmatrix} h'_{11} + h''_{11} & h'_{12} + h''_{12} \\ h'_{21} + h''_{21} & h'_{22} + h''_{22} \end{bmatrix} \begin{Bmatrix} i_1 \\ v_2 \end{Bmatrix}$$



4.2 互连网络

4.2.2 网络的并联

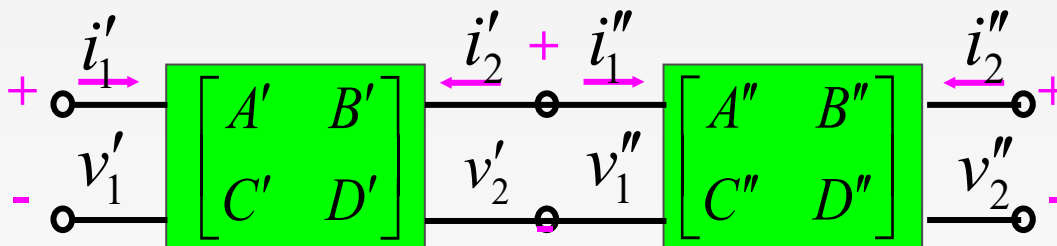
每个电流相互叠加而电压不变
则用Y参数:



$$\begin{Bmatrix} i_1 \\ i_2 \end{Bmatrix} = \begin{Bmatrix} i'_1 + i''_1 \\ i'_2 + i''_2 \end{Bmatrix} = \begin{bmatrix} Y'_{11} + Y''_{11} & Y'_{12} + Y''_{12} \\ Y'_{21} + Y''_{21} & Y'_{22} + Y''_{22} \end{bmatrix} \begin{Bmatrix} v_1 \\ v_2 \end{Bmatrix}$$

4.2 互联网络

4.2.3 级连网络



$$\begin{aligned} \begin{Bmatrix} v_1 \\ i_1 \end{Bmatrix} &= \begin{Bmatrix} v_1' \\ i_1' \end{Bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{Bmatrix} v_2' \\ -i_2' \end{Bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{Bmatrix} v_1'' \\ i_1'' \end{Bmatrix} \\ &= \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} A'' & B'' \\ C'' & D'' \end{bmatrix} \begin{Bmatrix} v_1'' \\ -i_2'' \end{Bmatrix} \end{aligned}$$

A参数特别适合级连网络

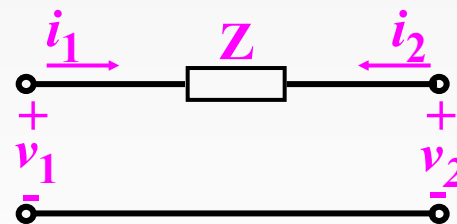
4.2 互连网络

4.2.3 级连网络

例4.4 求阻抗元件的ABCD参量。

解：

$$A = \left. \frac{v_1}{v_2} \right|_{i_2=0} = 1 \quad B = \left. \frac{v_1}{-i_2} \right|_{v_2=0} = Z$$
$$C = \left. \frac{i_1}{v_2} \right|_{i_2=0} = 0 \quad D = \left. \frac{i_1}{-i_2} \right|_{v_2=0} = 1$$



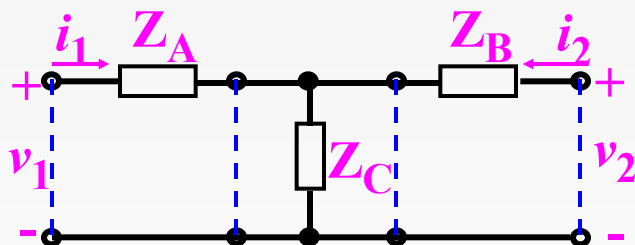
4.2 互连网络

4.2.3 级连网络

例4.5 求 T 形网络的ABCD参量。

解：

$$\begin{aligned} [ABCD] &= \begin{bmatrix} 1 & Z_A \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Z_C^{-1} & 1 \end{bmatrix} \begin{bmatrix} 1 & Z_B \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 + Z_A / Z_C & Z_A + Z_B + Z_A Z_B / Z_C \\ 1 / Z_C & 1 + Z_B / Z_C \end{bmatrix} \end{aligned}$$



4.2 互连网络

例4.6 求传输线段的A参量。

解： 当端口2 开路时，

$$V(d) = 2V^+ \cos(\beta d), \quad I(d) = \frac{2jV^+}{Z_0} \sin(\beta d)$$

当端口2 短路时，

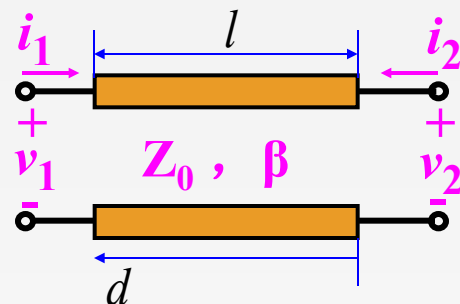
$$V(d) = 2jV^+ \sin(\beta d), \quad I(d) = \frac{2V^+}{Z_0} \cos(\beta d)$$

$$A = \frac{v_1}{v_2} \Big|_{i_2=0} = \frac{2V^+ \cos(\beta l)}{2V^+} = \cos(\beta l)$$

$$B = \frac{v_1}{-i_2} \Big|_{v_2=0} = \frac{2jV^+ \sin(\beta l)}{2V^+ / Z_0} = jZ_0 \sin(\beta l)$$

$$C = \frac{i_1}{v_2} \Big|_{i_2=0} = \frac{2jV^+ \sin(\beta l) / Z_0}{2V^+} = j \sin(\beta l) / Z_0$$

$$D = \frac{i_1}{-i_2} \Big|_{v_2=0} = \frac{2V^+ \cos(\beta l) / Z_0}{2V^+ / Z_0} = \cos(\beta l)$$



$I(d)$ 流向负载,
 $i_1 = I(d), i_2 = -I(d)$

4.3 网络特性及其应用

4.3.1 网络参量之间的换算关系

由h 参量方程 $\begin{cases} v_1 = h_{11}i_1 + h_{12}v_2 \\ i_2 = h_{21}i_1 + h_{22}v_2 \end{cases}$ 导出A参量:

$$A = \left. \frac{v_1}{v_2} \right|_{i_2=0} = \frac{h_{11}(-h_{22}v_2 / h_{21}) + h_{12}v_2}{v_2} = -\frac{h_{11}h_{22} - h_{12}h_{21}}{h_{21}} = -\frac{\Delta h}{h_{21}}$$

$$B = \left. \frac{v_1}{-i_2} \right|_{v_2=0} = \frac{h_{11}i_1}{-i_2} = \frac{h_{11}(i_2 / h_{21})}{-i_2} = -\frac{h_{11}}{h_{21}}$$

$$C = \left. \frac{i_1}{v_2} \right|_{i_2=0} = \frac{-h_{22}v_2 / h_{21}}{v_2} = -\frac{h_{22}}{h_{21}}$$

$$D = \left. \frac{i_1}{-i_2} \right|_{v_2=0} = \frac{i_2 / h_{21}}{-i_2} = -\frac{1}{h_{21}}$$

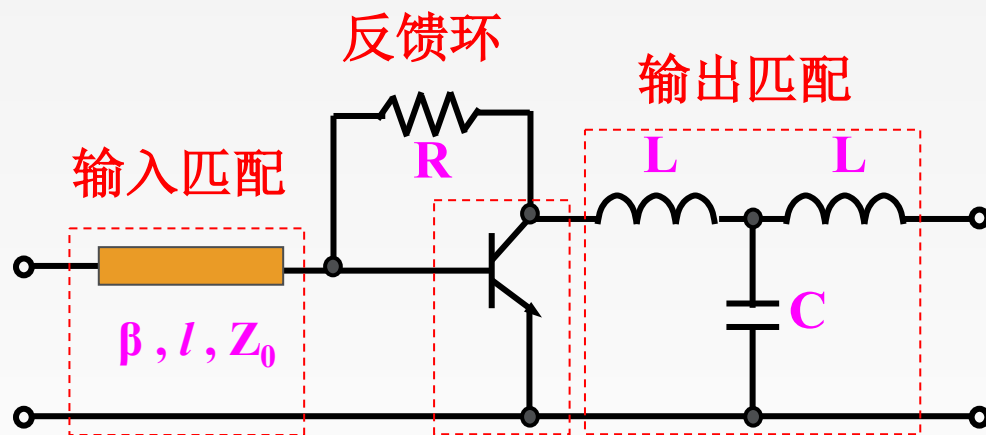
用同样方法可导出各参量之间的变换关系。

表 4.2 不同网络参量之间的变换关系

	[Z]	[Y]	[h]	[A]
[Z]	$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{Z_{22}}{\Delta Z} & -\frac{Z_{12}}{\Delta Z} \\ -\frac{Z_{21}}{\Delta Z} & \frac{Z_{11}}{\Delta Z} \end{bmatrix}$	$\begin{bmatrix} \frac{\Delta Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ -\frac{Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$	$\begin{bmatrix} \frac{Z_{11}}{Z_{21}} & \frac{\Delta Z}{Z_{21}} \\ \frac{1}{Z_{21}} & \frac{Z_{22}}{Z_{21}} \end{bmatrix}$
[Y]	$\begin{bmatrix} \frac{Y_{22}}{\Delta Y} & -\frac{Y_{12}}{\Delta Y} \\ -\frac{Y_{21}}{\Delta Y} & \frac{Y_{11}}{\Delta Y} \end{bmatrix}$	$\begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{Y_{11}} & -\frac{Y_{12}}{Y_{11}} \\ \frac{Y_{21}}{Y_{11}} & \frac{\Delta Y}{Y_{11}} \end{bmatrix}$	$\begin{bmatrix} \frac{Y_{22}}{Y_{21}} & -\frac{1}{Y_{21}} \\ -\frac{\Delta Y}{Y_{21}} & \frac{Y_{11}}{Y_{21}} \end{bmatrix}$
[h]	$\begin{bmatrix} \frac{\Delta h}{h_{22}} & \frac{h_{12}}{h_{22}} \\ -\frac{h_{21}}{h_{22}} & \frac{1}{h_{22}} \end{bmatrix}$	$\begin{bmatrix} \frac{1}{h_{11}} & -\frac{h_{12}}{h_{11}} \\ \frac{h_{21}}{h_{11}} & \frac{\Delta h}{h_{11}} \end{bmatrix}$	$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix}$	$\begin{bmatrix} -\frac{\Delta h}{h_{21}} & -\frac{h_{11}}{h_{21}} \\ \frac{h_{22}}{h_{21}} & -\frac{1}{h_{21}} \end{bmatrix}$
[A]	$\begin{bmatrix} \frac{A}{C} & \frac{\Delta ABCD}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix}$	$\begin{bmatrix} \frac{D}{B} & -\frac{\Delta ABCD}{B} \\ -\frac{1}{B} & \frac{A}{B} \end{bmatrix}$	$\begin{bmatrix} \frac{B}{D} & \frac{\Delta ABCD}{D} \\ -\frac{1}{D} & \frac{C}{D} \end{bmatrix}$	$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$

4.3 网络特性及其应用

4.3.2 微波放大器分析

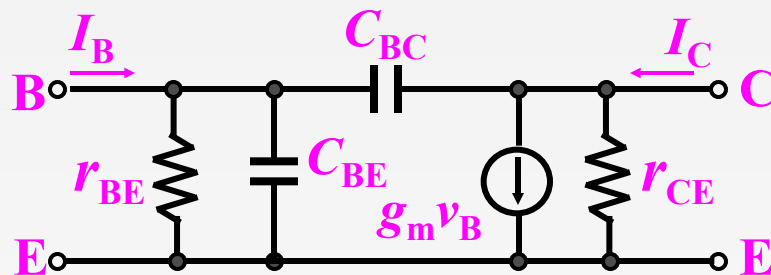


分析思路:

- 将 h 参量变换为 Y 参量与反馈环并联
- 变换为 A 参量与匹配网络级连。

4.3 网络特性及其应用

4.3.2 微波放大器分析



$$h_{11} = \frac{v_1}{i_1} \Big|_{v_2=0} = \frac{v_1}{v_1(1/r_{BE} + j\omega C_{BE} + j\omega C_{BC})} = \frac{r_{BE}}{1 + j\omega(C_{BE} + C_{BC})r_{BE}}$$

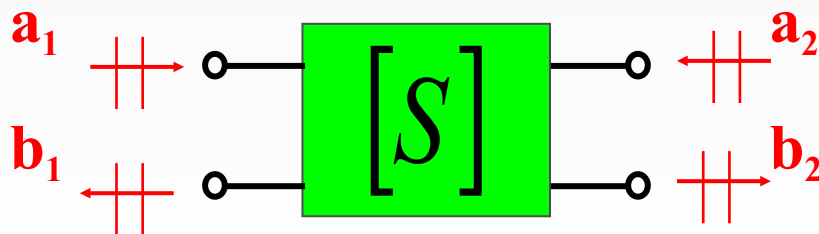
$$h_{12} = \frac{v_1}{v_2} \Big|_{i_1=0} = \frac{r_{BE} / (1 + j\omega r_{BE} C_{BE})}{1 / j\omega C_{BC} + r_{BE} / (1 + j\omega r_{BE} C_{BE})} = \frac{j\omega C_{BC} r_{BE}}{1 + j\omega(C_{BE} + C_{BC})r_{BE}}$$

$$h_{21} = \frac{i_2}{i_1} \Big|_{v_2=0} = \frac{g_m v_1 - j\omega C_{BC} v_1}{v_1(1/r_{BE} + j\omega C_{BE} + j\omega C_{BC})} = \frac{r_{BE}(g_m - j\omega C_{BC})}{1 + j\omega(C_{BE} + C_{BC})r_{BE}}$$

$$h_{22} = \frac{i_2}{v_2} \Big|_{i_1=0} = \frac{1}{r_{CE}} + \frac{g_m j\omega C_{BC} r_{BE}}{1 + j\omega(C_{BE} + C_{BC})r_{BE}} + \frac{j\omega C_{BC}(1 + j\omega C_{BE} r_{BE})}{1 + j\omega(C_{BE} + C_{BC})r_{BE}}$$

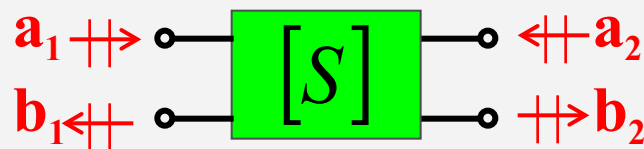
4.4 散射参量(S)

- 实际的射频系统不能采用终端开路(电容效应)或短路(电感效应)的测量方法,
- 终端的不连续性将导致有害的电压电流波反射, 并产生可能造成器件损坏的振荡。



4.4 散射参量(S)

4.4.1 S 参量的定义



定义S 参量:
$$\begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \end{Bmatrix}$$

其中:
$$S_{ij} = \left. \frac{b_i}{a_j} \right|_{a_n=0(n \neq j)}$$

定义归一化入射电压波: $a_n = \frac{V_n + Z_0 I_n}{2\sqrt{Z_0}}$

定义归一化反射电压波: $b_n = \frac{V_n - Z_0 I_n}{2\sqrt{Z_0}}$

相加: $V_n = (a_n + b_n)\sqrt{Z_0}$

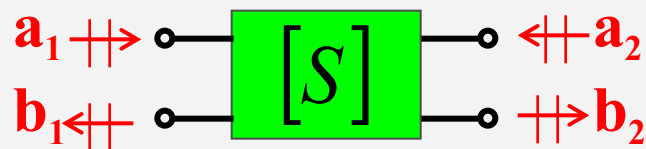
相减: $I_n = (a_n - b_n)/\sqrt{Z_0}$

所以:
$$a_n = V_n^+ / \sqrt{Z_0} = I_n^+ \sqrt{Z_0}, \quad b_n = V_n^- / \sqrt{Z_0} = -I_n^- \sqrt{Z_0}$$



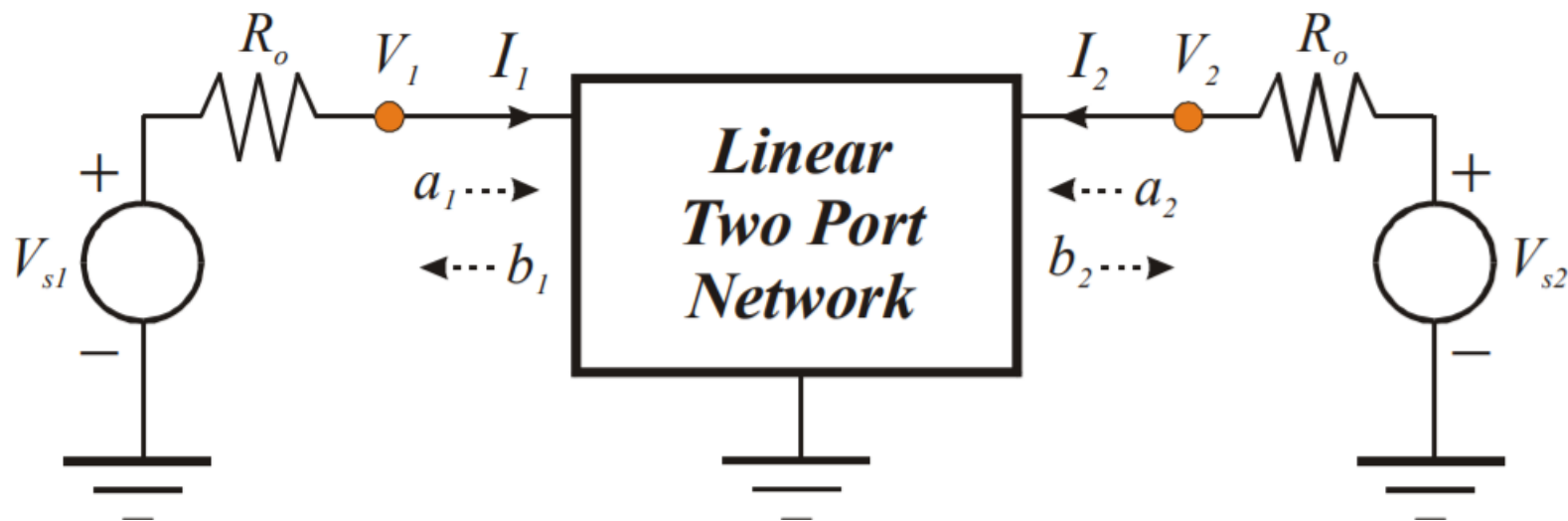
4.4 散射参量(S)

4.4.1 S 参量的定义



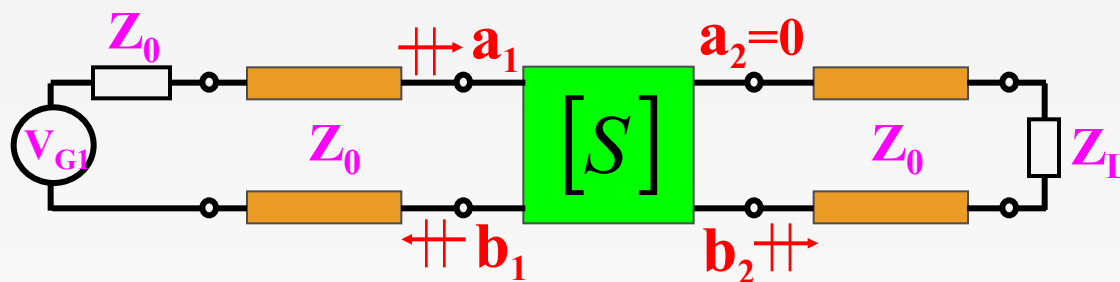
定义S 参量:
$$\begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \end{Bmatrix}$$

其中:
$$S_{ij} = \frac{b_i}{a_j} \Big|_{a_n=0(n \neq j)}$$



4.4 散射参量(S)

4.4.2 S 参量的物理意义



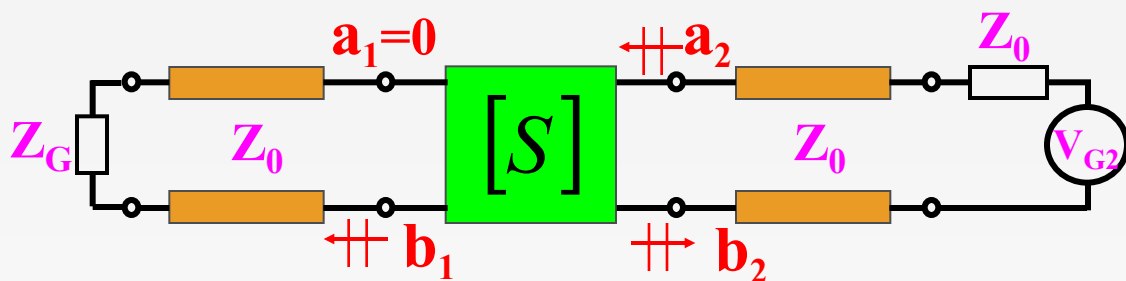
测量 S_{11} 和 S_{21} , 为保证 $a_2=0$, 必须使 $Z_L=Z_0$

$$\text{则: } S_{11} = \left. \frac{b_1}{a_1} \right|_{a_2=0} = \frac{V_1^-}{V_1^+} = \Gamma_{in} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0}$$

$$S_{21} = \left. \frac{b_2}{a_1} \right|_{a_2=0} = \frac{V_2^- / \sqrt{Z_0}}{(V_1 + Z_0 I_1) / (2\sqrt{Z_0})} \Big|_{I_2^+ = V_2^+ = 0} = \frac{2V_2^-}{V_{G1}} = \frac{2V_2}{V_{G1}} \quad \text{正向电压增益}$$

4.4 散射参量(S)

4.4.2 S 参量的物理意义



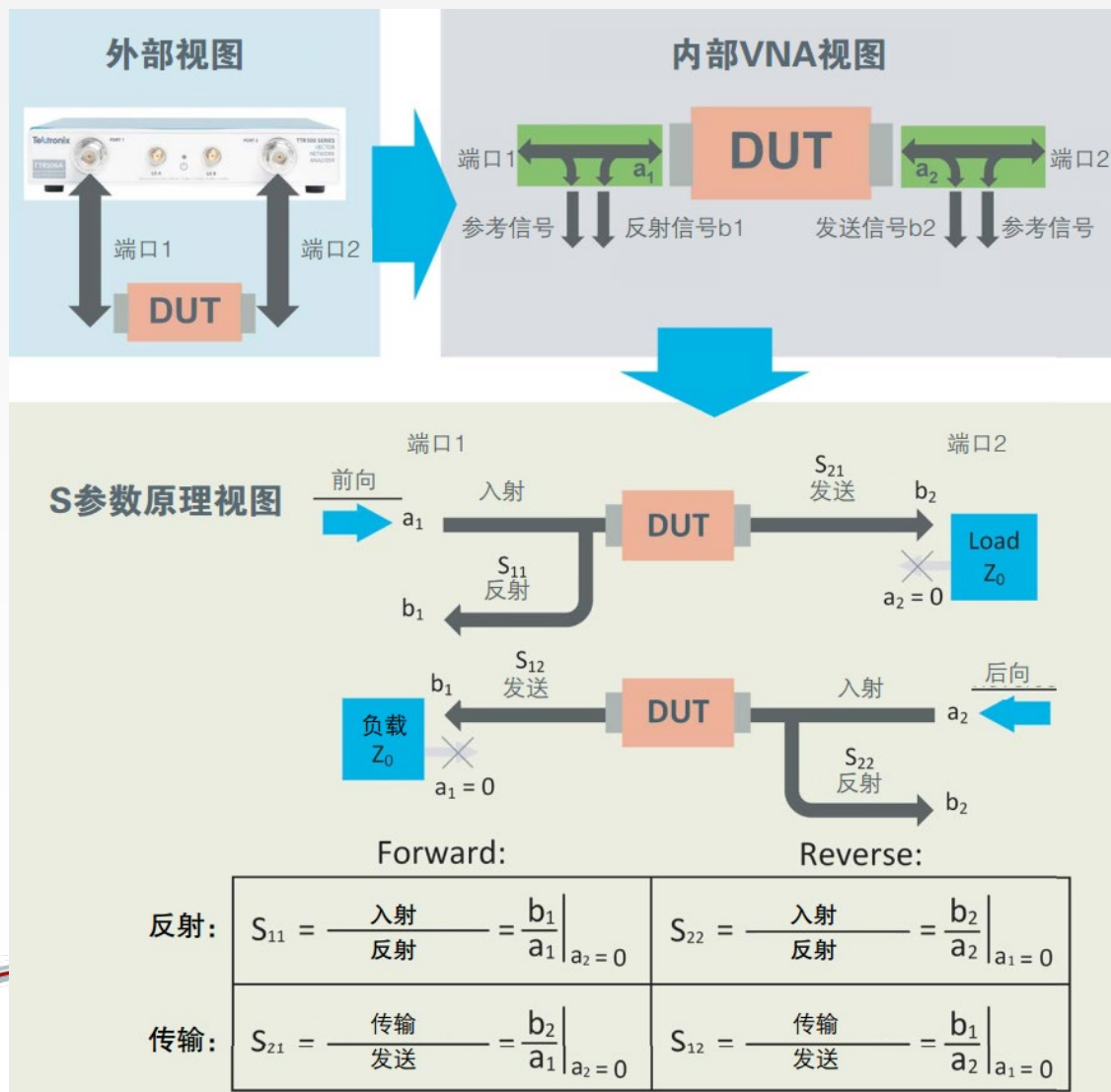
测量 S_{22} 和 S_{12} , 为保证 $a_1=0$, 必须使 $Z_G=Z_0$

$$\text{则: } S_{22} = \left. \frac{b_2}{a_2} \right|_{a_1=0} = \frac{V_2^-}{V_2^+} = \Gamma_{out} = \frac{Z_{out} - Z_0}{Z_{out} + Z_0}$$

$$S_{12} = \left. \frac{b_1}{a_2} \right|_{a_1=0} = \frac{V_1^- / \sqrt{Z_0}}{(V_2 + Z_0 I_2) / (2\sqrt{Z_0})} \Big|_{I_1^+ = V_1^+ = 0} = \frac{2V_1^-}{V_{G2}} = \frac{2V_1}{V_{G2}} \quad \text{反向电压增益}$$

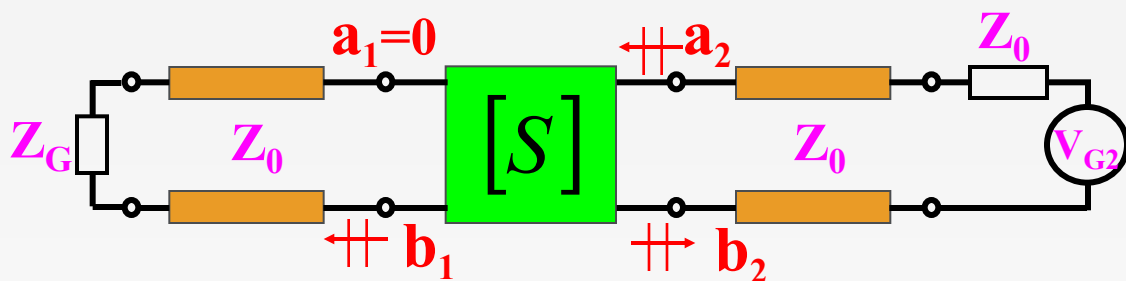
4.4 散射参量(S)

4.4.2 S 参量的物理意义



4.4 散射参量(S)

4.4.2 S 参量的物理意义



常用S参数名称

前向反射系数

- 输入回波损耗
- 输入匹配
- VSWR

S_{11}

前向传输系数

- 增益
- 损耗

S_{21}

反向传输系数

- 反向隔离度

S_{12}

反向反射系数

- 输出回波损耗
- 输出匹配
- VSWR

S_{22}

4.4 散射参量(S)

4.4.2 S 参量的物理意义

增益与S参数关系

网络增益与S参数之间关系如下

$$G = S_{21} = \frac{b_2}{a_1}$$

取标量可得

$$|G| = |S_{21}|$$

取对数表示则如下

$$g = 20 \log_{10} |S_{21}| \text{ dB}$$

	Forward:	Reverse:
反射:	$S_{11} = \frac{\text{入射}}{\text{反射}} = \frac{b_1}{a_1} \Big _{a_2=0}$	$S_{22} = \frac{\text{入射}}{\text{反射}} = \frac{b_2}{a_2} \Big _{a_1=0}$
传输:	$S_{21} = \frac{\text{传输}}{\text{发送}} = \frac{b_2}{a_1} \Big _{a_2=0}$	$S_{12} = \frac{\text{传输}}{\text{发送}} = \frac{b_1}{a_2} \Big _{a_1=0}$

4.4 散射参量(S)

4.4.2 S 参量的物理意义

电压驻波比 (VSWR) 与S参数关系

电压驻波比表示在端接任意负载的情况下，传输线Z₀上可以测量到的最大电压与最小电压之比（驻波波峰电压与波谷电压的比值）。

对于输入端口 VSWR (s_{in}) 表示为

$$s_{in} = \frac{1 + |S_{11}|}{1 - |S_{11}|}$$

对于输出端口VSWR (s_{out}) 表示为

$$s_{out} = \frac{1 + |S_{22}|}{1 - |S_{22}|}$$

	Forward:	Reverse:
反射:	$S_{11} = \frac{\text{入射}}{\text{反射}} = \frac{b_1}{a_1} \Big _{a_2=0}$	$S_{22} = \frac{\text{入射}}{\text{反射}} = \frac{b_2}{a_2} \Big _{a_1=0}$
传输:	$S_{21} = \frac{\text{传输}}{\text{发送}} = \frac{b_2}{a_1} \Big _{a_2=0}$	$S_{12} = \frac{\text{传输}}{\text{发送}} = \frac{b_1}{a_2} \Big _{a_1=0}$

4.4 散射参量(S)

4.4.2 S 参量的物理意义

回波损耗（Return Loss）与S参数关系

$$RL(\text{dB}) = 10 \log_{10} \frac{P_i}{P_r}$$

其中 P_i 为输入功率, P_r 为反射功率。

对于二端口网络输入端口回波损耗为

$$RL_{\text{in}} = 10 \log_{10} \left| \frac{1}{S_{11}^2} \right| = -20 \log_{10} |S_{11}|$$

输出端口回波损耗为

$$RL_{\text{out}} = -20 \log_{10} |S_{22}| \text{ dB}$$

	Forward:	Reverse:
反射:	$S_{11} = \frac{\text{入射}}{\text{反射}} = \frac{b_1}{a_1} \Big _{a_2=0}$	$S_{22} = \frac{\text{入射}}{\text{反射}} = \frac{b_2}{a_2} \Big _{a_1=0}$
传输:	$S_{21} = \frac{\text{传输}}{\text{发送}} = \frac{b_2}{a_1} \Big _{a_2=0}$	$S_{12} = \frac{\text{传输}}{\text{发送}} = \frac{b_1}{a_2} \Big _{a_1=0}$

4.4 散射参量(S)

4.4.2 S 参量的物理意义

插入损耗（**Insert Loss**）与**S**参数关系

插入损耗指在传输系统的某处由于元件或器件的插入而发生的负载功率的损耗

$$IL = -20 \log_{10} |S_{21}| \text{ dB.}$$

	Forward:	Reverse:
反射:	$S_{11} = \frac{\text{入射}}{\text{反射}} = \frac{b_1}{a_1} \Big _{a_2=0}$	$S_{22} = \frac{\text{入射}}{\text{反射}} = \frac{b_2}{a_2} \Big _{a_1=0}$
传输:	$S_{21} = \frac{\text{传输}}{\text{发送}} = \frac{b_2}{a_1} \Big _{a_2=0}$	$S_{12} = \frac{\text{传输}}{\text{发送}} = \frac{b_1}{a_2} \Big _{a_1=0}$

4.4 散射参量(S)

例4.7 假设一3dB衰减网络插入到 $Z_0 = 50\Omega$ 的传输线中，求该网络的S参量和电阻。

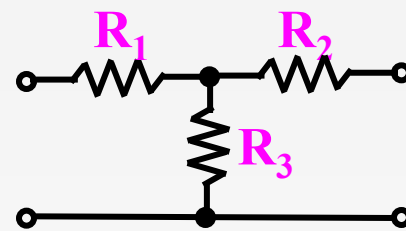
解： 由于网络必须匹配，所以： $S_{11} = S_{22} = 0$

则：
$$Z_{in} = R_1 + \frac{R_3(R_2 + 50)}{R_3 + R_2 + 50} = 50\Omega$$

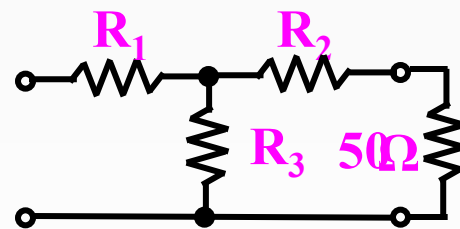
$$V_2 = \left(\frac{\frac{R_3(R_2 + 50)}{R_3 + R_2 + 50}}{R_1 + \frac{R_3(R_2 + 50)}{R_3 + R_2 + 50}} \right) \frac{50V_1}{R_2 + 50}$$

在3dB衰减条件下：
$$S_{21} = \frac{2V_2}{V_{G1}} = \frac{V_2}{V_1} = 0.707 = S_{12}$$

根据对称性联立得： $R_1 = R_2 = 8.58\ \Omega$ ， $R_3 = 141.4\ \Omega$



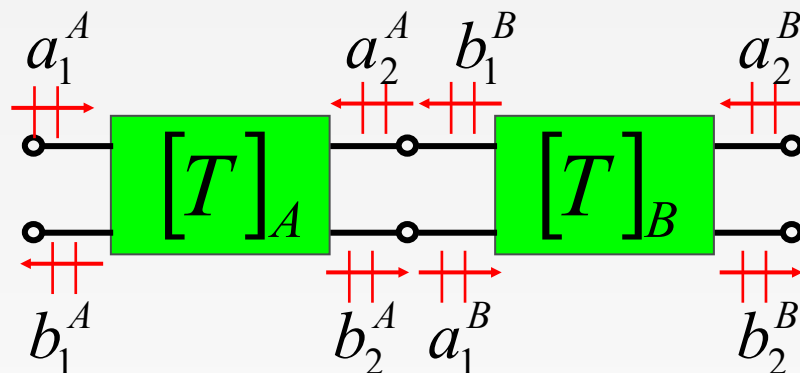
衰减网络



S_{11} 和 S_{21} 测试电路

4.4 散射参量(S)

4.4.3 链形S 矩阵



最有效的方法是按输入输出口分类重写电压波关系式：

$$\begin{Bmatrix} a_1^A \\ b_1^A \end{Bmatrix} = \begin{bmatrix} T_{11}^A & T_{12}^A \\ T_{21}^A & T_{22}^A \end{bmatrix} \begin{bmatrix} T_{11}^B & T_{12}^B \\ T_{21}^B & T_{22}^B \end{bmatrix} \begin{Bmatrix} b_2^B \\ a_2^B \end{Bmatrix}$$

链形散射矩阵与
A矩阵作用相同

4.4 散射参量(S)

4.4.4 Z 参量与S 参量之间的转换

$$\begin{aligned} \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} &= Z_0 \begin{bmatrix} 1 + S_{11} & S_{12} \\ S_{21} & 1 + S_{22} \end{bmatrix} \begin{bmatrix} 1 - S_{11} & -S_{12} \\ -S_{21} & 1 - S_{22} \end{bmatrix}^{-1} \\ &= \frac{Z_0 \begin{bmatrix} 1 + S_{11} & S_{12} \\ S_{21} & 1 + S_{22} \end{bmatrix}}{(1 - S_{11})(1 - S_{22}) - S_{21}S_{12}} \begin{bmatrix} 1 - S_{22} & S_{12} \\ S_{21} & 1 - S_{11} \end{bmatrix} \end{aligned}$$

4.4 散射参量(S)

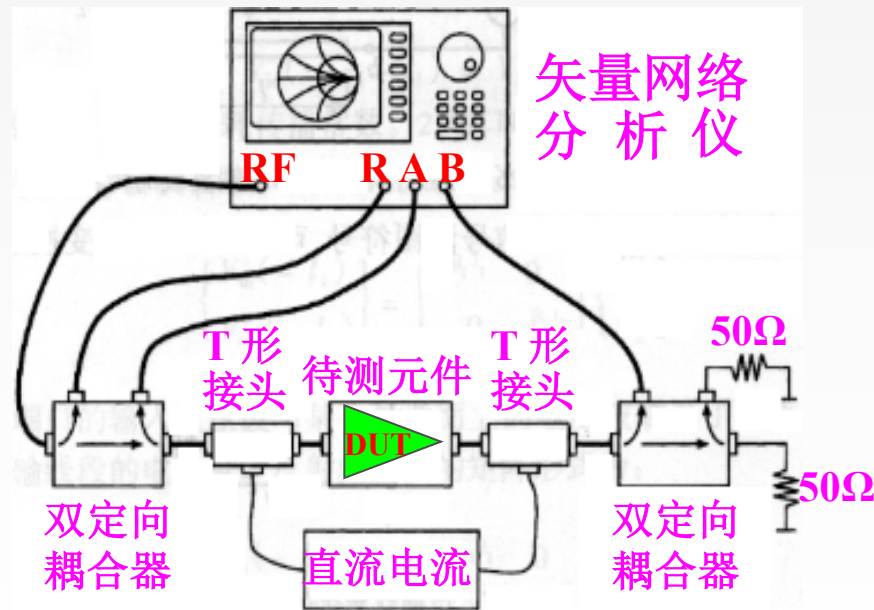
4.4.4 Z 参量与S 参量之间的转换

矩阵参量	用[s]表示	用[z]表示	用[y]表示	用[ā]表示
[s]	$\begin{bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{bmatrix}$	$[s] = ([z] + [1])^{-1} \times [z] - [1]$ $s_{11} = \frac{ z + z_{11} - z_{22} - 1}{ z + z_{11} + z_{22} + 1}$ $s_{12} = \frac{2z_{12}}{ z + z_{11} + z_{22} + 1}$ $s_{21} = \frac{2z_{21}}{ z + z_{11} + z_{22} + 1}$ $s_{22} = \frac{ z - z_{11} + z_{22} - 1}{ z + z_{11} + z_{22} + 1}$	$[s] = ([1] + [y])^{-1} \times [1] - [y]$ $s_{11} = \frac{1 - y_{11} + y_{22} - y }{1 + y_{11} + y_{22} + y }$ $s_{12} = \frac{-2y_{12}}{1 + y_{11} + y_{22} + y }$ $s_{21} = \frac{-2y_{21}}{1 + y_{11} + y_{22} + y }$ $s_{22} = \frac{1 + y_{11} - y_{22} - y }{1 + y_{11} + y_{22} + y }$	$s_{11} = \frac{\bar{a} + \bar{b} - \bar{c} - \bar{d}}{\bar{a} + \bar{b} + \bar{c} + \bar{d}}$ $s_{12} = \frac{2 \bar{a} }{\bar{a} + \bar{b} + \bar{c} + \bar{d}}$ $s_{21} = \frac{2}{\bar{a} + \bar{b} + \bar{c} + \bar{d}}$ $s_{22} = \frac{-\bar{a} + \bar{b} - \bar{c} + \bar{d}}{\bar{a} + \bar{b} + \bar{c} + \bar{d}}$
[z]	$[z] = ([1] - [s])^{-1} \times ([1] + [s])$ $z_{11} = \frac{1 + s_{11} - s_{22} - s }{1 - s_{11} - s_{22} + s }$ $z_{12} = \frac{2s_{12}}{1 - s_{11} - s_{22} + s }$ $z_{21} = \frac{2s_{21}}{1 - s_{11} - s_{22} + s }$ $z_{22} = \frac{1 - s_{11} + s_{22} - s }{1 - s_{11} - s_{22} + s }$	$[z] = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$	$[z] = [y]^{-1}$ $= \frac{1}{ y } \times \begin{bmatrix} y_{22} & -y_{12} \\ -y_{21} & y_{11} \end{bmatrix}$	$[z] = \frac{1}{\bar{c}} \begin{bmatrix} \bar{a} & \bar{a} \\ 1 & \bar{d} \end{bmatrix}$

4.4 散射参量(S)

4.4.6 S 参量的测量

- 射频源RF输出射频信号，
- 测量通道R用于测量入射波，同时也作为参考端口。
- 通道A和B:测量反射波和传输波($S_{11}=A/R$, $S_{21}=B/R$)。
- 若要测量 S_{12} 和 S_{22} ，将待测元件反接。



测量 S_{11} 和 S_{21} 的实验系统

4.4 散射参量(S)

4.4.6 S 参量的测量

为什么要进行校正测量

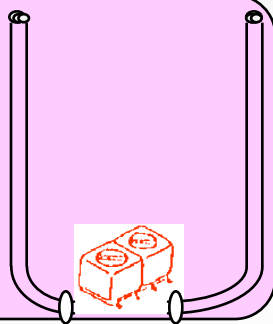
A =

待测元件特性



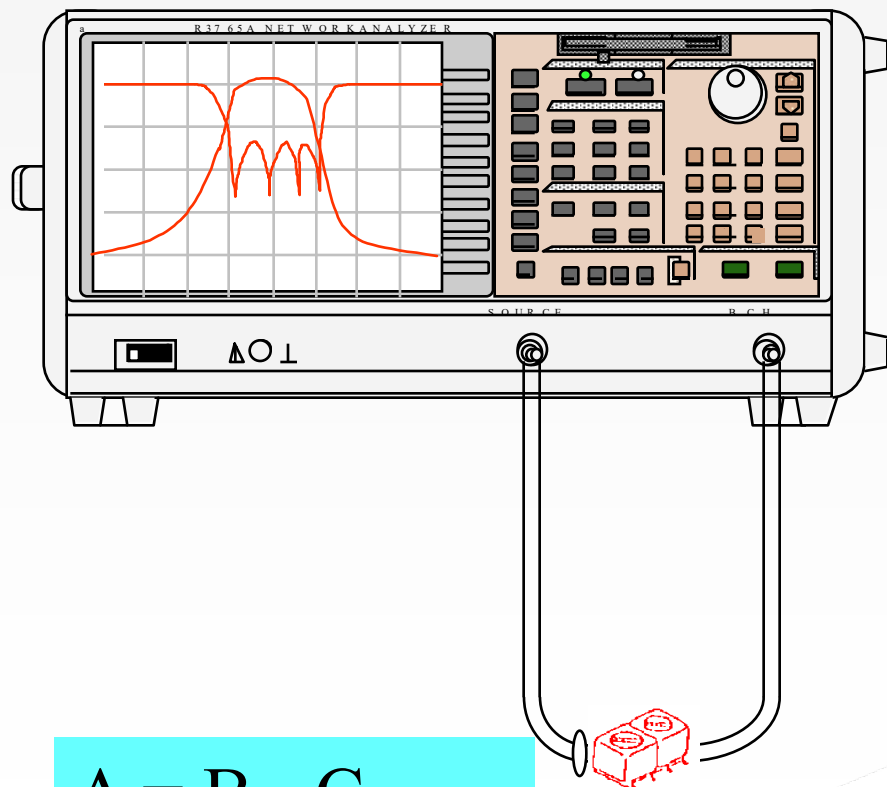
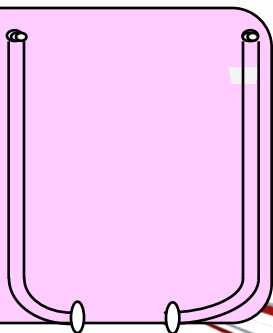
B =

全体的特性



C =

测定系统的特性

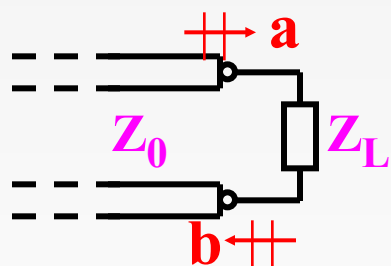


$$A = B - C$$

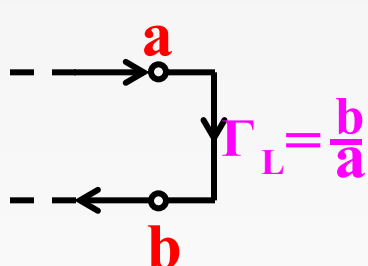
4.4.5 信号流图模型

主要原则：

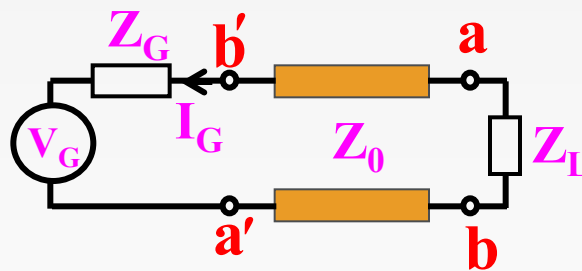
1. 当涉及S参量时，节点是用来识别网络参量的；
2. 支路是用来连接网络参量的；
3. 支路量值的加减与支路的走向有关。



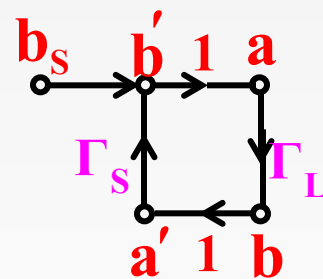
常规形式



信号流图



常规形式



信号流图

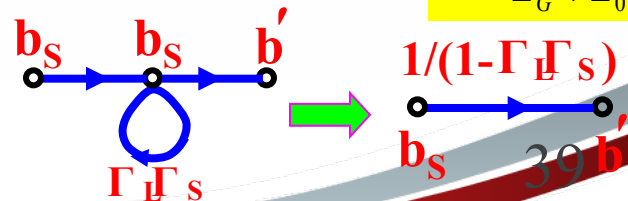
根据相加原则： $b' = b_s + a'\Gamma_S$ ，所以： $b_s = b' - a'\Gamma_S$

考虑到信号源： $V_S = V_S^+ + V_S^- = V_G + Z_G I_G = V_G + Z_G \left(\frac{V_S^+}{Z_0} - \frac{V_S^-}{Z_0} \right)$

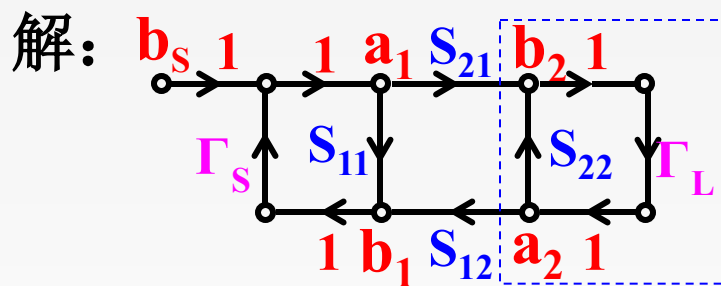
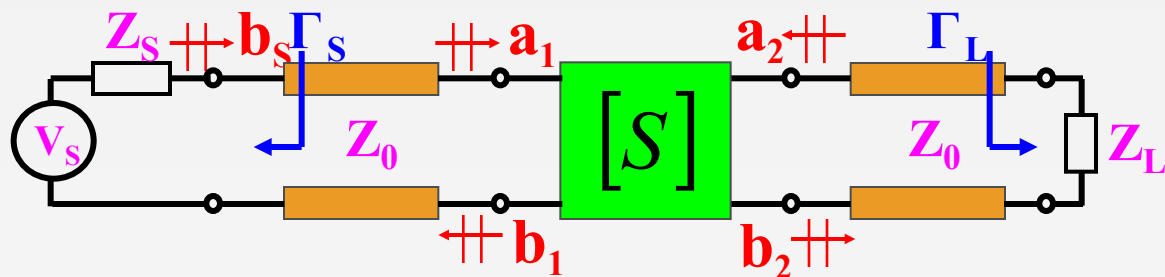
整理并同除 $\sqrt{Z_0}$ ： $\frac{\sqrt{Z_0}}{Z_G + Z_0} V_G = \frac{V_S^-}{\sqrt{Z_0}} - \Gamma_S \frac{V_S^+}{\sqrt{Z_0}} = b' - a'\Gamma_S = b_s$

$$\Gamma_S = \frac{Z_G - Z_0}{Z_G + Z_0}$$

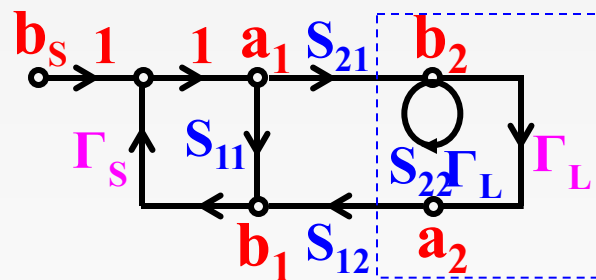
重要结论： $b' = b_s + \Gamma_L \Gamma_S b' = \frac{b_s}{1 - \Gamma_L \Gamma_S}$



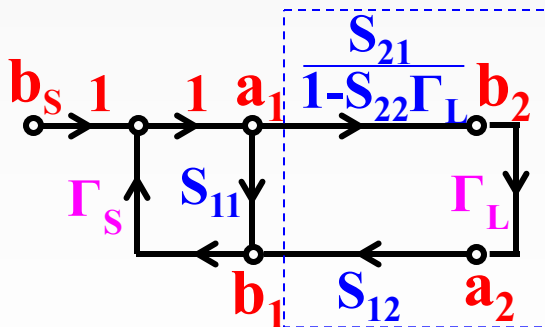
例4.8 求图示网络中的 b_1/a_1 和 a_1/b_s ，传输线倍乘因子为1。



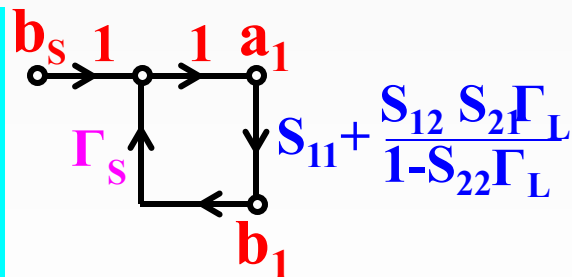
1. 断开 b_2 与 a_2 之间的环路并形成反馈环 $S_{22}\Gamma_L$



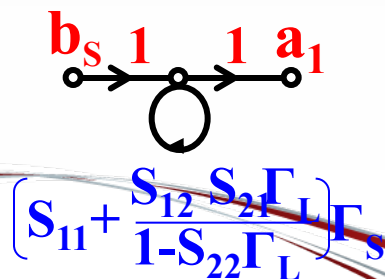
2. 分解 a_1 与 b_2 之间的反馈环化为 $S_{21}/(1 - S_{22}\Gamma_L)$



3. 完成串并联运算并求 Γ_{in}



4. 将环路变为反馈环，求倍乘因子



5. 分解反馈环最后得

$$a_1 = \frac{1}{1 - \left[S_{11} + \frac{S_{12} S_{21} \Gamma_L}{1 - S_{22} \Gamma_L} \right] \Gamma_s} b_s$$

4.4.6 S 参量的推广

如果与端口1、2相连接的传输线特性阻抗不相同，假设分别为 Z_{01} 、 Z_{02}

则：

$$V_n = V_n^+ + V_n^- = (a_n + b_n)\sqrt{Z_{0n}},$$

$$I_n = V_n^+ / Z_{0n} - V_n^- / Z_{0n} = (a_n - b_n) / \sqrt{Z_{0n}}$$

故：

$$a_n = \frac{V_n^+}{\sqrt{Z_{0n}}}, \quad b_n = \frac{V_n^-}{\sqrt{Z_{0n}}},$$

$$S_{ij} = \left. \frac{b_i}{a_j} \right|_{a_n=0(n \neq j)} = \left. \frac{V_i^- / \sqrt{Z_{0i}}}{V_j^+ / \sqrt{Z_{0j}}} \right|_{V_n^+=0(n \neq j)}$$

4.4.6 S 参量的推广

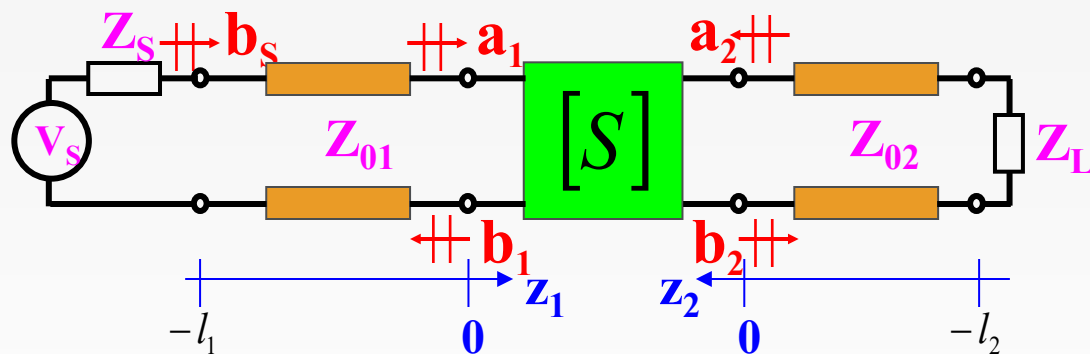
考虑测量网络S参量时需要利用一段有限长的传输线，对端口1：

则： $V_{in}^+(z_1 = 0) = V_1^+$

$$V_{in}^+(z_1 = -l_1) = V_1^+ e^{-j\beta_1(-l_1)}$$

$$V_{in}^-(z_1 = 0) = V_1^-$$

$$V_{in}^-(z_1 = -l_1) = V_1^- e^{j\beta_1(-l_1)}$$



端口 2 同理，则

$$\begin{Bmatrix} V_{in}^+(-l_1) \\ V_{out}^+(-l_2) \end{Bmatrix} = \begin{bmatrix} e^{j\beta_1 l_1} & 0 \\ 0 & e^{j\beta_2 l_2} \end{bmatrix} \begin{Bmatrix} V_1^+ \\ V_2^+ \end{Bmatrix}$$

$$\begin{Bmatrix} V_{in}^-(-l_1) \\ V_{out}^-(-l_2) \end{Bmatrix} = \begin{bmatrix} e^{-j\beta_1 l_1} & 0 \\ 0 & e^{-j\beta_2 l_2} \end{bmatrix} \begin{Bmatrix} V_1^- \\ V_2^- \end{Bmatrix}$$

思考题

4.1 求T形网络的阻抗矩阵和导纳矩阵.

4.2 根据 $ABCD$ 参量的定义求 Y 参量矩阵.

4.4 已知传输线的特性阻抗为 75Ω , 终端接 25Ω 负载, 求回波损耗.

4.5 已知放大器输入、输出端口的驻波系数分别为 $VSWR=2$ 和 $VSWR=3$, 求输入、输出端口反射系数的模. 若采用 S_{11} 和 S_{22} 表示计算结果, 其物理含义是什么?

